

CDN Judo: Breaking the CDN DoS Protection with Itself

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Run Guo (Tsinghua University), Weizhong Li (Tsinghua University), Baojun Liu (Tsinghua University), Shuang Hao (University of Texas at Dallas), Jia Zhang (Tsinghua University), Haixin Duan (Tsinghua University), Kaiwen Sheng (Tsinghua University), Jianjun Chen (ICSI), Ying Liu (Tsinghua University)

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Content Delivery Network (CDN) improves the websites' accessing performance and availability with its globally distributed network infrastructures, which contributes to the flourish of CDN-powered websites on the Internet. As CDN-powered websites are normally operating important businesses or critical services, the attackers are mostly interested to take down these high-value websites, achieving severe damage with maximum influence. As the CDN absorbs distributed attacking traffic with its massive bandwidth resources, CDN vendors have always claimed that they provide effective DoS protection for the CDN-powered websites.

However, we reveal that, implementation or protocol weaknesses in the CDN's forwarding mechanism can be exploited to break the CDN protection. By sending crafted but legal requests, an attacker can launch an efficient DoS attack against the website Origin behind. In particular, we present three CDN threats in this study. Through abusing the CDN's HTTP/2 request converting behavior and HTTP pre-POST behavior, an attacker can saturate the CDN-Origin bandwidth and exhaust the Origin's connection limits. What is more concerning is that, some CDN vendors only use a small set of traffic forwarding IPs with lower IP-churning ratio to establish connections with the Origin. This characteristic provides a great opportunity for an attacker to effectively degrade the website's global availability, by just cutting off specific CDN-Origin connections.

In this work, we examine the CDN's request-forwarding behaviors across six well-known CDN vendors, and we perform real-world experiments to evaluate the severity of the threats. As the threats are caused by the CDN vendor's poor trade-offs between usability and security, we discuss the possible mitigations, and we receive positive feedback after responsible disclosure to related CDN vendors.

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